

ID:

Name:

**Brac University**

Semester: Fall 2025

Course Code: CSE250

Circuits And Electronics

Section: 02

Faculty: PDS



Assessment: Quiz 2

Duration: 20 minutes

Date: November 18, 2025

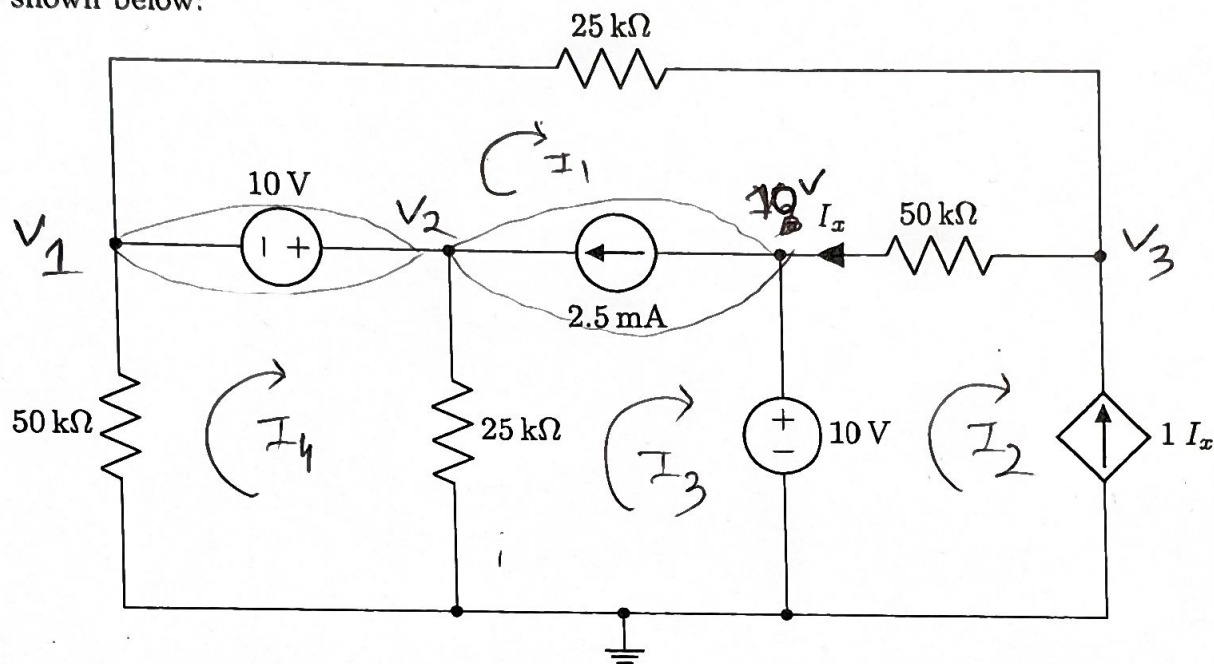
Full Marks (incl. bonus 3): 23

- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ All 1 question(s) are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Bonus questions are indicated as "(bonus)" along with allotted marks.
- ✓ Write your answers inside the indicated boxes where applicable.
- ✓ Symbols have their usual meanings.

■ Question 1 of 1

[CO3] [20 marks + 3 marks (bonus)]

Consider the circuit shown below:



- (a) [3 marks (bonus)] The circuit has the following three mesh equations, where I_1 , I_2 , I_3 , and I_4 are the mesh currents (in mA):

$$I_2 = -1 I_x \quad (1)$$

$$I_1 - I_3 = 2.5 \quad (2)$$

$$50 I_4 - 10 + 25 (I_4 - I_3) = 0 \quad (3)$$

Derive the fourth independent equation required to solve the circuit using Mesh Analysis. You do not need to simplify or solve the equations.

KVL at I_1 & I_3 ,

$$25(I_3 - I_4) + 10 + 25I_1 + 50(I_1 - I_2) + 10 = 0$$

- (b) [20 marks] Apply Nodal Analysis to determine the power of the 10 V source on the left.

KVL at the supernode,

$$V_1 - V_2 = -10 \quad \text{--- (i)}$$

$$\text{KCL at } V_1 \text{ \& } V_3, \quad \frac{V_1}{50} + \frac{V_1 - V_3}{25} + \frac{V_2}{25} = 2.5$$

$$\Rightarrow 3V_1 + 2V_2 - 2V_3 = 125 \quad \text{--- (ii)}$$

$$\Rightarrow 2V_1$$

KCL at V_3 ,

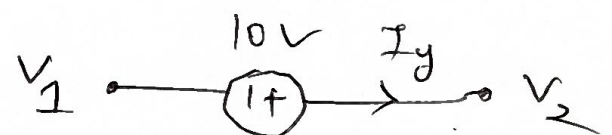
$$1I_x = \underbrace{\frac{V_3 - 10}{50}}_{I_x} + \frac{V_3 - V_1}{25}$$

$$\Rightarrow V_3 = V_1$$

$$\Rightarrow V_1 - V_3 = 0 \quad \text{--- (iii)}$$

$$\text{solving ... , } V_1 = 35V ; V_2 = 45V ; V_3 = 35V$$

Let the current supplied by the 10V source is I_y



KCL at V_2 ,

$$I_y + 2.5 = \frac{V_2 - 0}{25}$$

$$\Rightarrow I_y = -0.7mA$$

$$P_{10V} = -10 \times I_y = 7mW$$

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Inspiring Excellence

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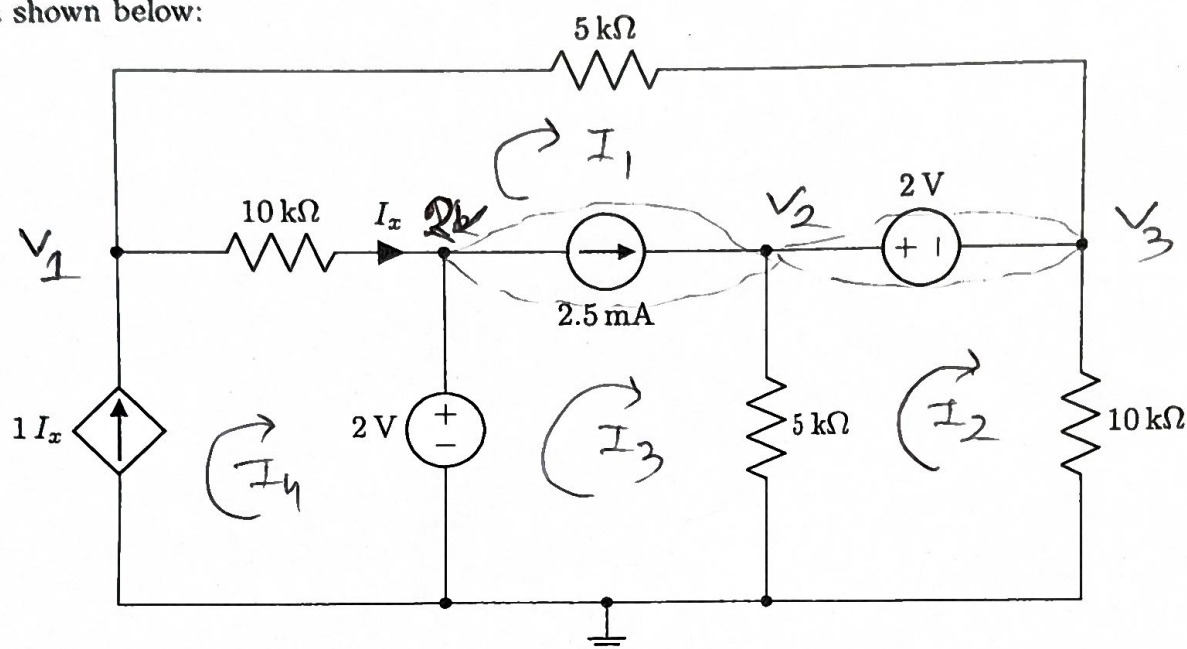
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■ Question 1 of 1

[CO3] [20 marks + 3 marks (bonus)]

Consider the circuit shown below:



- (a) [3 marks (bonus)] The circuit has the following three mesh equations, where I_1 , I_2 , I_3 , and I_4 are the mesh currents (in mA):

$$I_4 = 1 I_x \quad (4)$$

$$I_3 - I_1 = 2.5 \quad (5)$$

$$5(I_2 - I_3) + 2 + 10 I_2 = 0 \quad (6)$$

Derive the fourth independent equation required to solve the circuit using Mesh Analysis. You do not need to simplify or solve the equations.

KVL at I_1 & I_3 ,

$$-2 + 10(I_1 - I_4) + 5I_1 - 2 + 5(I_3 - I_2) = 0$$

- (b) [20 marks] Apply Nodal Analysis to determine the power of the 2V source on the right.

KVL at the supernode, $V_2 - V_3 = 2$ — (i)

KCL at V_2 & V_3 ,

$$2.5 = \frac{V_2 - 0}{5} + \frac{V_3 - 0}{10} + \frac{V_3 - V_1}{5} \quad \left| \quad 2V_1 + 2V_2 - 3V_3 = 25 \right. \quad \text{--- (ii)}$$

KCL at v_1 ,

$$1I_x = \underbrace{\frac{v_1 - 2}{10}}_{I_x} + \frac{v_1 - v_3}{5}$$

$$\Rightarrow v_1 - v_3 = 0 \text{ — (iii)}$$

solving ..., $v_1 = 7V$; $v_2 = 9V$; $v_3 = 7V$

Let the current supplied by the 2V source is I_y ,

KCL at v_2 ,

$$2.5 + I_y = \frac{v_2 - 0}{5}$$

$$\Rightarrow I_y = -0.7 \text{ mA}$$

$$P_{2V} = -2 \times I_y = 1.4 \text{ mW}$$

